

Advanced RAG Systems: Technical Specifications

1. Executive Summary

This document outlines the architecture for a next-generation Retrieval-Augmented Generation (RAG) system. The goal is to maximize retrieval precision while maintaining a low-latency generation loop using Groq inference.

We focus on modularity, enabling the swapping of embedding models and vector stores without disrupting the core pipeline logic.

2. System Architecture

- Ingestion Layer: Handles unstructured data parsing.
- Vector Layer: Multi-modal embedding and semantic indexing.
- Reasoning Layer: LangGraph-driven stateful agents.
- Evaluation Layer: Ragas-based metrics for faithfulness and relevance.